WEEK 2 TASK REPORT

Hands-on Practice in Mechanical Design

Internship: Mechanical Design Intern
Organization: YuvaIntern
Software Used: SOLIDWORKS

1. Objective

The objective of this exercise was to practice mechanical component modeling using SOLIDWORKS, apply engineering dimensions, understand the basics of Geometric Dimensioning and Tolerancing (GD&T), and prepare a technical drawing of a mechanical component.

2. Introduction

For this exercise, I created a 3D model of a flanged mounting bracket using SOLIDWORKS. A mounting bracket is a mechanical component used to support or connect parts and secure them in a fixed position. The model includes mounting surfaces and holes for fastening.

This exercise helped me understand the practical application of 3D modeling tools, engineering dimensions, and basic design considerations used in mechanical engineering.

3. Software and Tools Used

- SOLIDWORKS for 3D part modeling.
- Sketch tools for creating 2D profiles.
- Extruded Boss/Base for creating solid features.
- Extruded Cut for creating holes and cutouts.
- Fillet for rounding selected edges.
- Smart Dimension for defining sketch dimensions.
- Drawing tools for preparing technical views.

4. Modeling Procedure

Step 1: Create a New Part

I opened SOLIDWORKS and created a new part document. I selected the appropriate reference plane to begin the sketch.

Step 2: Create the Base Sketch

I drew the initial profile of the flanged mounting bracket using sketch tools. I applied dimensions to define the required shape and size.

Step 3: Create the 3D Model

I used the Extruded Boss/Base feature to give thickness to the sketch and create the main solid body of the bracket.

Step 4: Add Mounting Holes

I created sketches for the mounting holes and used the Extruded Cut feature to remove material and form the holes required for fastening.

Step 5: Apply Fillets

I applied the Fillet feature to selected edges to improve the component's geometry and remove sharp edges where appropriate.

Step 6: Verify the Model

I checked the overall shape, dimensions, and hole locations to ensure that the model matched my intended design.

5. Engineering Dimensions and GD&T Considerations

Dimensions are essential for defining the size and geometry of a mechanical component. In this exercise, dimensions were used to define the bracket profile, thickness, hole diameters, and hole positions.

Geometric Dimensioning and Tolerancing (GD&T) is used in engineering drawings to communicate acceptable geometric variation and help ensure proper assembly and functionality.

For this bracket, the following GD&T considerations are relevant:

- Datum reference: A suitable flat mounting surface can serve as a primary datum for inspection and positioning.
- Position tolerance: A positional tolerance may be specified for the mounting holes to control their locations relative to the selected datums.
- Flatness: A flatness tolerance may be applied to a mounting surface when required by the design.
- Dimensional tolerances: Appropriate tolerances should be specified according to functional requirements and manufacturing capability.

These tolerances should be selected based on actual design and manufacturing requirements rather than assigned arbitrarily.

6. Technical Drawing

A technical drawing can be prepared from the completed 3D model using the SOLIDWORKS Drawing environment. Front, top, and side views can be included to communicate the component's geometry clearly.

The drawing should show the important dimensions, hole diameters, hole locations, material information if specified, and applicable tolerances. Appropriate datum symbols and GD&T feature control frames can be added when required by the design.

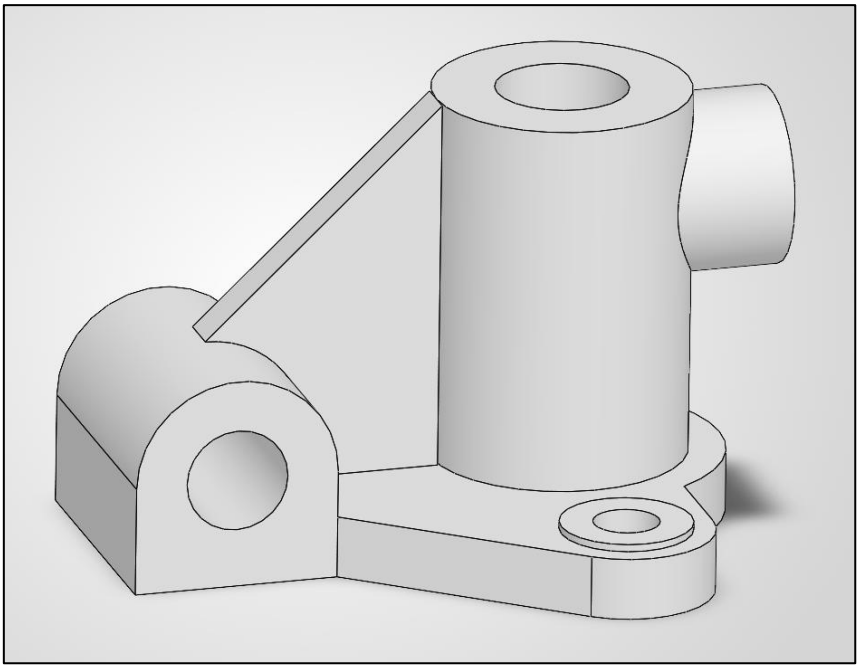


Fig 1: Flanged Mounting Bracket

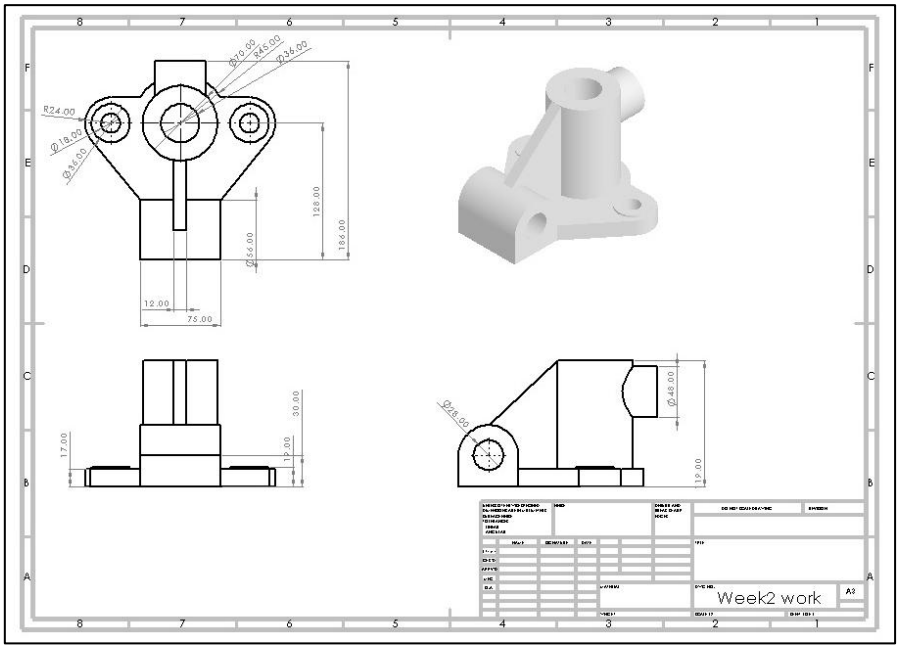


Fig 2: Technical Drawing with Dimensions and Views

7. Best Practices Followed

- Used sketches and dimensions to define the component geometry.
- Applied suitable modeling features to create the required shape.
- Checked important dimensions and hole locations.
- Considered datum selection and geometric tolerances for functional requirements.
- Focused on producing a clear and understandable mechanical model.

8. Conclusion

Through this exercise, I gained practical experience in creating a 3D mechanical component using SOLIDWORKS. I practiced sketching, extrusion, cutting, filleting, and dimensioning. I also developed a better understanding of the importance of GD&T, datum selection, and technical drawings in mechanical design.

This exercise helped me connect 3D modeling concepts with engineering drawing practices and strengthened my understanding of the mechanical design process.